

4

5

4

5

G.O

B&L

Trapizoid

Trapizoid

Trapizoid

Square

Prox.

Trapizoid

Trapizoid

Rhomboid

Rhomboid

occl.

hexagonal

oval

diamond

2-Cusp₂
oval
3-Cusp₂
square

Contact.A

M

M $\frac{1}{3}$

O $\frac{1}{3}$

D

more
occlusally

more
Cervically

(M & D)

M $\frac{1}{3}$

more Cervically

To (J) of

O & M $\frac{1}{3}$

as 4

Ch.F

Canine
Fossa

M.M.D.G



Snake eye

mesi-lingual
D.G

max.
coverity
at O $\frac{1}{3}$
Lingually

have 2 Roots

only one
root

Transvers R.

sever lingual
Convergency

L-Cusp $\frac{2}{3}$

B-Cusp

Short notes

[M] mesial

[D] Distal

[B] Buccal

[L] Lingual

[M.R] marginal Ridge

[D.G] develop mental groove

فكرة الرموز

upper 4

upper 5

Lobes

4 Lobe (3 buccal & 1 lingual)

4 Lobe (3 buccal & 1 lingual)

Contact Area

[M] middle 1/3

[D] more occlusally ↓

[M] occlusal 1/3

[D] more cervically ↑

Buccal Ridge

Prominent

Less prominent

Root

• Shorter

الوحيد اللي فيه ②

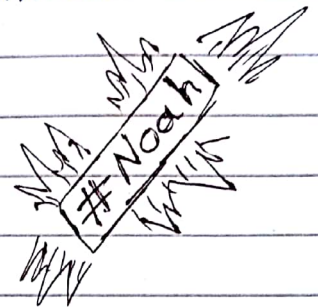
• Two Roots

85% والباقي واحد

• [L] Root < [B] Root

• Longer

• one root



Cusp

• (lingually) مقل كونه

• [M] slope > [D] slope

• [B] Cusp > [L] Cusp by 1mm

• [D] slope > [M] slope

• at the same level

Cervical Portion

narrow



wide



Occlusal Table

wide



less



Occlusal aspect

• hexagonal

• [M&D] [M.R] less thick

• Long central [D.G]

• oval

• [M&D] [M.R] Thicker

• short central [D.G]

Characteristic Features of 4

① mesial marginal Developmental groove [MMDG]

② mesial Developmental depression. (Canine fossa)

4

شوية ملاحظات مهمة في 4 ويمكن كمان موجودة بعضها في 5

Elevation

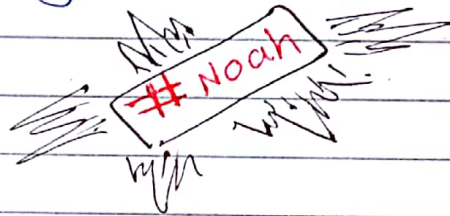
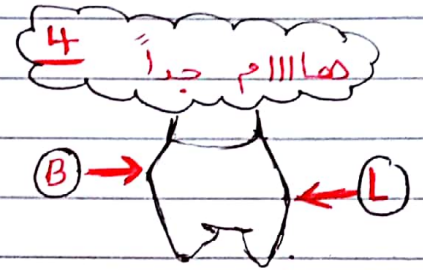
- Cervical Ridge
- Buccal Ridge
- Buccal Triangular Ridge
- Lingual Triangular Ridge
- (M&D) marginal Ridge

Depression

- shallow 2 depressions (M&D) to Buccal Ridge.
- mesial marginal developmental groove
- Central developmental groove
- (m & D) Triangular fossae
- mesial developmental depression

Buccal max. Convexity at $C \frac{1}{3}$

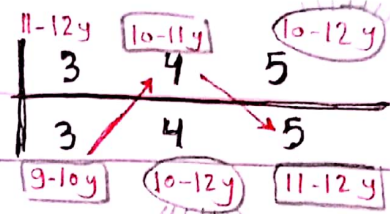
Lingual max. Convexity at $m \frac{1}{3}$



Transverse Ridge (L-Cusp $\frac{2}{3}$ B-Cusp) (Snake eye M D)

mandibular first premolar $\overline{4}$

→ 4 lobes (3 buccally, 1 lingually)



Geometric outline - Buccal & lingual → Trapezoid ▭

- proximally → Rhomboid ◇

- Occlusally → diamond ◇

Characteristic features

① mesio-lingual developmental groove

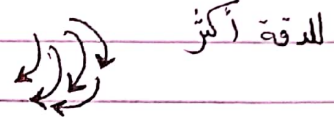
② Transverse ridge between B & L Cusp

③ M & D fossae (snake eye) M D

Facially → D slope > M slope

→ Contact Area: M & D contact areas → M $\frac{1}{3}$

* Cervically to the junction of O & M $\frac{1}{3}$



Lingually → Sever lingual convergence

→ Lingual Cusp $\frac{2}{3}$ the length of buccal Cusp

Root

→ Tapered مسطوية [apex] pointed مبدية curved distally

Elevation → (Cervical ridge): max. convexity at buccal aspect, C $\frac{1}{3}$

→ max. convexity at lingual aspect, M $\frac{1}{3}$

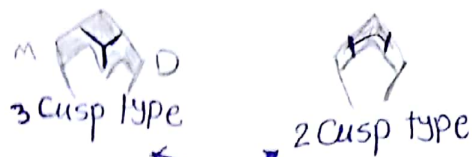
→ Buccal ridge → B & L triangular ridge → M & D marginal ridge

Depression → 2 M & D depression beside buccal ridge

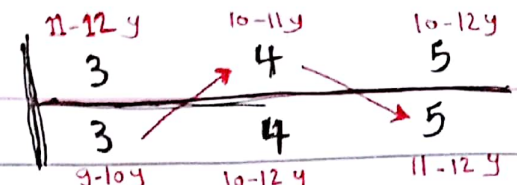
→ Central Developmental groove

→ M & D fossae

→ mesio-lingual developmental groove



mandibular second premolar 5



- no. Lobes**
- 2-Cusp type: 4 Lobes (3 buccal & 1 lingual)
 - 3-Cusp type: 5 Lobes (3 buccal & 2 lingual)

G. outline → Buccal & Lingual: square

→ Proximal: Rhomboid

→ occlusal: 2-cusp: oval
3-cusp: square

Lingual

2-Cusp: L-Cusp (shorter & smaller) than B-Cusp

3-Cusp: B-Cusp > mesio-lingual cusp > Disto-lingual Cusp
it have L-developmental groove between the two Cusps.

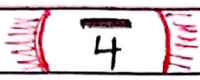
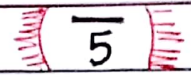
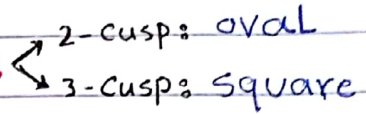
Elevations 2 & 3 Cusp: ① B-ridge ② Cervical ridge
2-cusp → ③ B & L triangular ridge ④ Transverse ridge
⑤ M & D marginal ridge

3-Cusp (only): mesio-lingual > Disto-lingual > triangular ridge

Depression ① M & D triangular fossae
② M & D groove beside B-ridge

③ Central developmental groove → 2-cusp: H or U shaped
3-cusp: y shaped

Ch. feature (lingually) max. convexity at $\frac{0}{3}$

	 4	 5
No. Lobes	4 lobes (3B & 1L)	2-Cusp: 4 lobes (3B & 1L) 3-Cusp: 5 lobes (3B & 2L)
G. outline	B & L: Trapezoid Proximal: Rhomboid occlusal: Diamond	B & L: square proximal: Rhomboid occlusal:  2-cusp: oval 3-cusp: square
B-Cusp	prominent	⊗ Short ⊗ less prominent
Root	⊗ short ⊗ pointed apex	⊗ Long ⊗ blunt apex
max. Convixty	 C $\frac{1}{3}$  M $\frac{1}{3}$	 C $\frac{1}{3}$  O $\frac{1}{3}$
L-inclination	prominent	less prominent
Marginal Ridges	 oblique /  straight —	 straight —  straight —
Fossae	 oval ○  circular ○ 	 Triangular  Triangular
Central D.G	Straight	 2-Cusp: <u>U</u> or <u>H</u> shaped  3-Cusp: <u>Y</u> shaped